Information Theory Workshop (ITW) 2026

# Information-theoretic foundations of anytime-valid inference

## Part 2 (11:00 am to 12:30pm)

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# Tutorial outline

**PART 1:**

*Introduction + finite alphabet methods*

**1. Motivation**

*A/B testing example; failure of classical methods; Anytime-valid (AV) testing*

**2. Fundamental Limits**

*Classical DPI + Wald arguments*

**3. AV Test for Finite Alphabets**

*Universal Inference; Proof of optimality*

**PART 2:**

*General Principles*

**1. Bounded mean testing**

*Testing-by-betting principle, Kelly criterion, universal portfolios.*

**2. Unifying Design Principle**

*Donsker-Varadhan representation, Likelihood-free Universal Inference.*

**3. Extensions**

*Sequential estimation, change detection, non-iid data, etc.*

# Recap: the sequential testing problem

- Sequential testing problem on $\mathcal{X}$:

$$X_1, X_2, \ldots \overset{\text{iid}}{\sim} P.$$

$$H_0 : P_X \in \mathcal{P}_0 \qquad \text{vs.} \qquad H_1 : P_X \in \mathcal{P}_1.$$

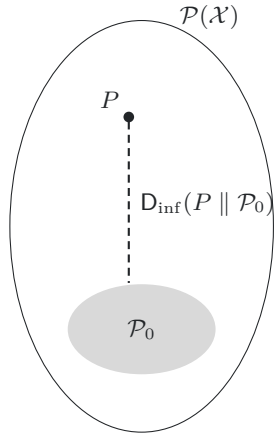
- Design a stopping time $\tau_\alpha$ at which we reject $H_0$, with

$$\underbrace{\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha < \infty) \leq \alpha,}_{\text{Validity under } H_0} \qquad \underbrace{\inf_{P \in \mathcal{P}_1} \mathbb{P}_P(\tau_\alpha < \infty) = 1\,.}_{\text{“Power-one” under } H_1}$$

- Any such $\tau_\alpha$ must satisfy

$$\mathbb{E}_P[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)},$$

where $\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) := \inf_{Q \in \mathcal{P}_0} D(P\|Q)$.




**Goal:** Design $\tau_\alpha$ that matches this limit as $\alpha \downarrow 0$.

# Recap: universal inference

- In Part 1, we focused on $\mathcal{X} = \{1, \ldots, k\}$.

- **General approach:**
  (i) Define a process $\{E_t : t \geq 1\}$ that quantifies evidence against $H_0$.
  (ii) Reject $H_0$ when $E_t$ reaches the threshold $1/\alpha$.

- In Part 1, we used a mixture-to-maximum likelihood ratio process:

$$E_t = \frac{\displaystyle\int_{\Delta_{k-1}} \prod_{i=1}^{t} q(X_i)\, \pi(dq)}{\displaystyle\sup_{Q \in \mathcal{P}_0} \prod_{i=1}^{t} Q(X_i)} = \frac{\text{mixture likelihood}}{\text{maximum null likelihood}}.$$

- However, in many problems, a full likelihood is unavailable or difficult to compute.

# A nonparametric example: bounded-mean testing

- **Observations:** $\mathcal{X} = [0,1]$, and $X_1, X_2, \ldots \overset{\text{iid}}{\sim} P$
- **Testing problem:** For some fixed $m \in (0,1)$, decide b/w

$$H_0 : \mathbb{E}_P[X] \leq m, \quad \text{versus} \quad H_0 : \mathbb{E}_P[X] > m.$$

- The null posits that the mean of this stream equals $m$; equivalently,

$$\mathcal{P}_0 = \big\{P \in \mathcal{P}([0,1]) : \mathbb{E}_P[X] \leq m\big\}.$$

- **There is no common $\sigma$-finite dominating measure** for $\mathcal{P}_0$.
- The same is true for many other problems; e.g., *two-sample testing, independence testing etc.*

# The principle of testing by betting

*Evidence against a null hypothesis can be formally quantified as the gain in wealth of a fictitious bettor repeatedly betting on observed outcomes, at odds that are fair or subfair under $H_0$.*

**Initialize:** $W_0 \leftarrow 1$.

**For** $t = 1, 2, \ldots$ **do:**

1. **Place a bet:** Choose a fraction $u_t \in [0, 1]$ of current wealth to bet, and the payoff rule.
2. **Observe** the next outcome $X_t$, and collect the payoff.
3. **Update** the bettor's wealth $W_t$.

**End for.**

# Designing a (sub-)fair bet for $H_0 : \mathbb{E}[X_t] \leq m$

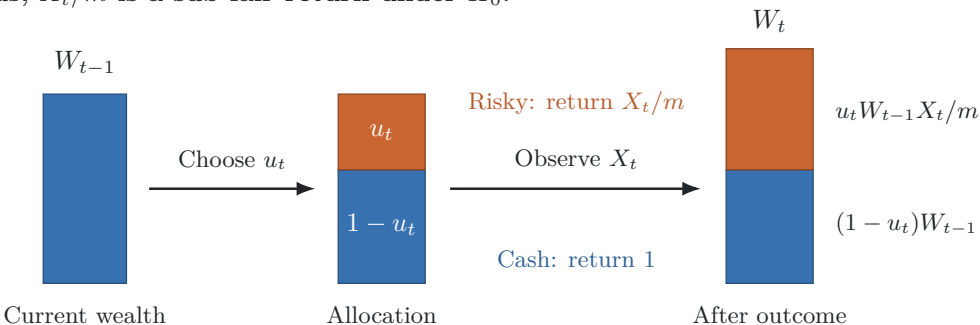
$$\mathbb{E}_Q[X_t] \leq m \quad \Longrightarrow \quad \mathbb{E}_Q\left[\frac{X_t}{m}\right] \leq 1, \quad \text{for all } Q \in \mathcal{P}_0.$$

**Thus, $X_t/m$ is a sub-fair return under $H_0$.**

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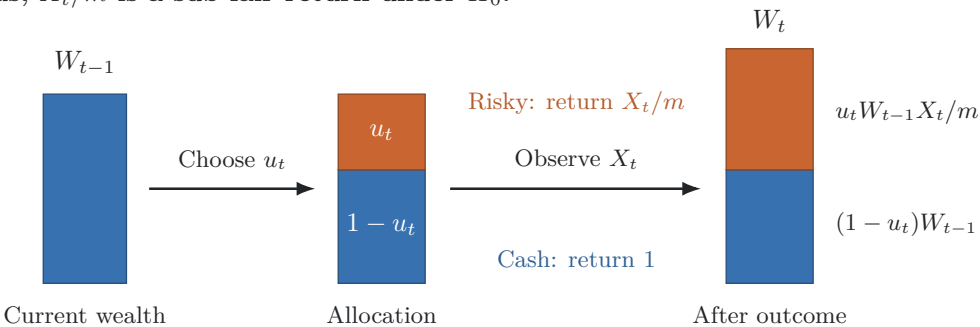
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**Thus, $X_t/m$ is a sub-fair return under $H_0$.**




$$W_t = W_{t-1}\left((1-u_t) + u_t\frac{X_t}{m}\right), \qquad 0 \le u_t \le 1.$$

# Testing by betting for bounded means

Testing $H_0 : \mathbb{E}[X] \leq m$, with $X_t \in [0,1]$.

**Initialize:** $W_0 = 1$.

**For** $t = 1, 2, \ldots$:

1. **Choose** $u_t \in [0,1]$ using only $X^{t-1}$.
2. **Observe** $X_t$.
3. **Update** $W_t = W_{t-1}\big(1 - u_t + u_t X_t / m\big)$.
4. **Reject** $H_0$ and stop if $W_t \geq 1/\alpha$.

$$\tau_\alpha = \inf\{t \geq 1 : W_t \geq 1/\alpha\}$$

How should we choose the sequence of bets $\{u_t\}$?

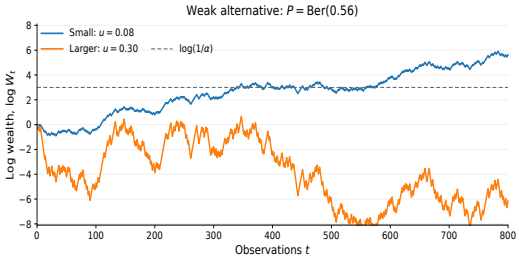
# Testing by betting for bounded means

**Under** $H_0$**:** $X_t \sim \mathrm{Ber}(0.50)$. Ten independent runs.

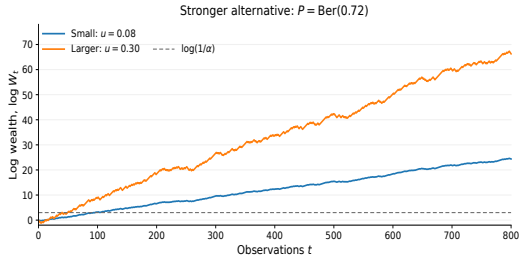



Under the null, no bet can gain money consistently.

# Testing by betting for bounded means




The smaller bet works well.




The larger bet works well.

Under the alternative, different bets might perform better.

# Any betting strategy works under $H_0$

Recall that $u_t \in [0,1]$ is chosen using only $\mathcal{F}_{t-1} = \sigma(X_1, \ldots, X_{t-1})$, and

$$W_t = W_{t-1}\left(1 - u_t + u_t \frac{X_t}{m}\right), \qquad W_0 = 1.$$

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Under any $Q \in \mathcal{P}_0$,

$$\mathbb{E}_Q[W_t \mid \mathcal{F}_{t-1}] = W_{t-1}\left(1 - u_t + u_t \frac{\mathbb{E}_Q[X_t]}{m}\right) \leq W_{t-1}.$$

Thus, $\{W_t\}_{t \geq 0}$ is a **nonnegative supermartingale** under $H_0$.

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By Ville's inequality, we have

$$\sup_{Q \in \mathcal{P}_{0,m}} \mathbb{P}_Q\left(\sup_{t\ge 0} W_t \ge \frac{1}{\alpha}\right) \le \alpha, \quad \text{hence} \quad \tau_\alpha = \inf\left\{t \ge 1 : W_t \ge \frac{1}{\alpha}\right\} \text{ is valid for } \mathcal{P}_0$$

Ville. *Étude critique de la notion de collectif* (1939)

# How should we bet?

Suppose we use a fixed $u \in [0, 1]$ in every round:

$$W_n(u) = \prod_{t=1}^{n} \left(1 - u + u\frac{X_t}{m}\right).$$

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Under an i.i.d. alternative $P$, by the SLLN,

$$\begin{aligned}\frac{1}{n}\log W_n(u) &= \frac{1}{n}\sum_{t=1}^{n} \log\left(1 - u + u\frac{X_t}{m}\right) \\ &\xrightarrow{\text{a.s.}} \underbrace{\mathbb{E}_P\left[\log\left(1 - u + u\frac{X}{m}\right)\right]}_{=:g_P(u)}.\end{aligned}$$

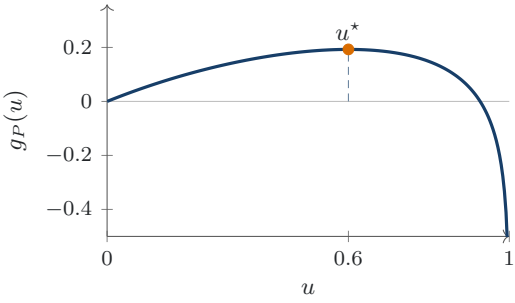
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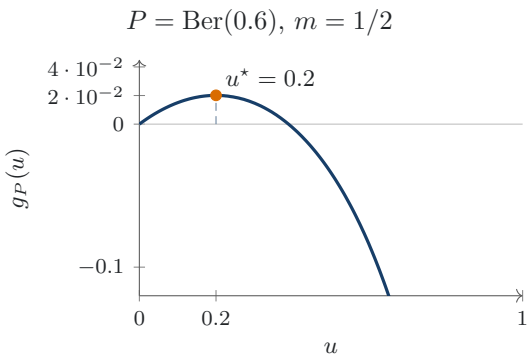


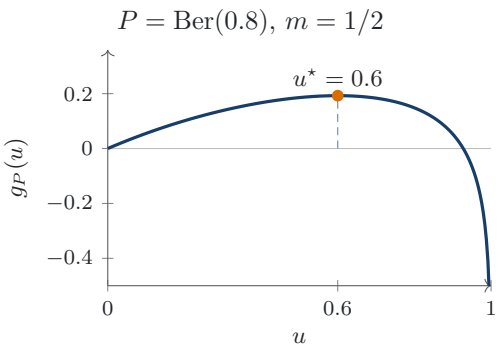

The **log-optimal or Kelly bet** is

$$u^\star \in \arg\max_{u \in [0,1]} g_P(u).$$

# Learning the Kelly bet by mixing

**Challenge.** The Kelly bet $u^\star$ depends on the unknown alternative $P$.



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**Idea.** Instead of committing to a single $u$, mix over all constant allocations $u \in [0,1]$.

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**Validity.** Mixing preserves the nonnegative supermartingale property under every $Q \in \mathcal{P}_0$.

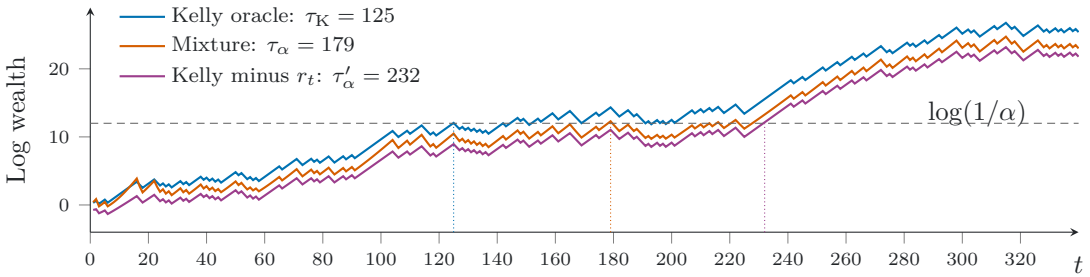
**Efficiency?** How much log-wealth does the mixture lose relative to the Kelly bet $u^\star$?

# From portfolio regret to stopping time

For mixing distribution $\pi = \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$, we have a pathwise guarantee

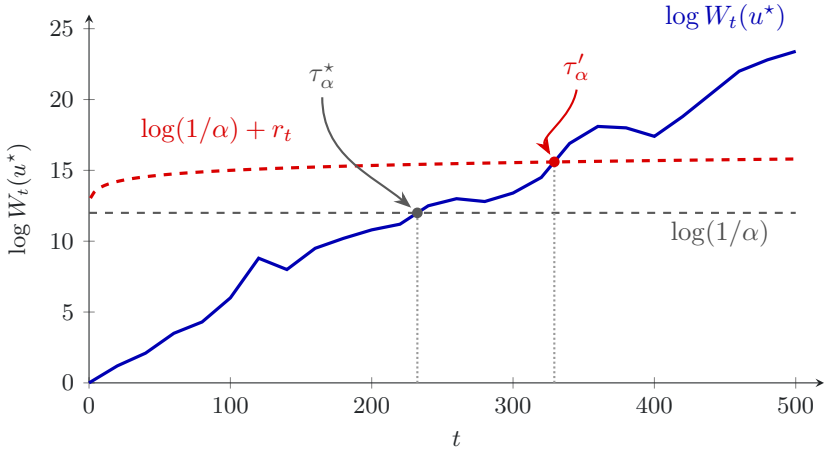
$$\log \overline{W}_t \geq \sup_{u \in [0,1]} \log W_t(u) - r_t, \qquad r_t = \tfrac{1}{2}\log(t+1) + \log 2.$$

Hence, $\tau_\alpha \leq \tau'_\alpha := \inf\{t \geq 1 : \log W_t(u^\star) - r_t \geq \log(1/\alpha)\}$.



# From portfolio regret to stopping time

$$\tau'_\alpha = \inf\left\{t \geq 1 : \log W_t(u^\star) \geq \log(1/\alpha) + r_t\right\}, \qquad r_t = \frac{1}{2}\log(t+1) + \log 2.$$



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By the regret bound, we have $\tau_\alpha \leq \tau'_\alpha$.

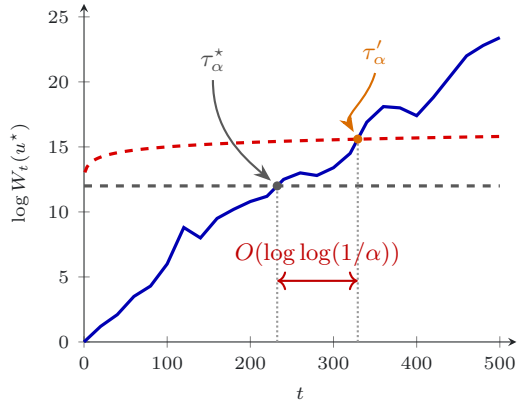
Under $P$, the oracle process grows linearly:

$$\frac{1}{t}\log W_t(u^\star) \to g_P^\star > 0.$$

Hence, $\tau_\alpha$ is finite a.s. (**power one**).

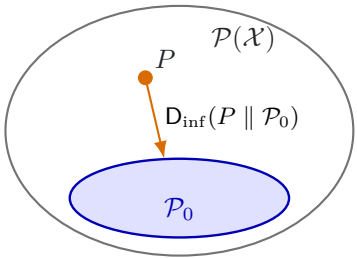
Since $\tau_\alpha \leq \tau'_\alpha$, we also have

$$\mathbb{E}_P[\tau_\alpha] \leq \frac{\log(1/\alpha) + O(\log\log(1/\alpha))}{g_P^\star}.$$



# Is this test optimal?

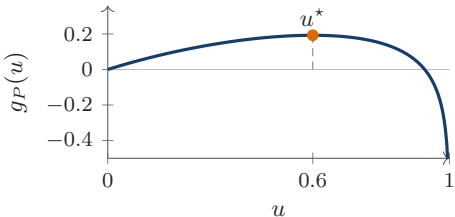
**Fundamental limit (Part 1)**




Every level-$\alpha$, power-one test satisfies

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**Testing-by-betting**

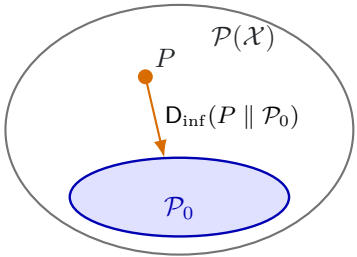



The proposed test $\tau_\alpha$ achieves

$$\mathbb{E}_P[\tau_\alpha] \leq \frac{\log(1/\alpha)}{g_P^\star}\,(1 + o(1))\,.$$

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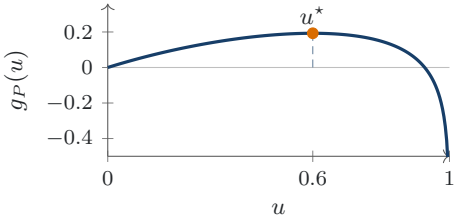
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**Is the Kelly growth rate equal to the minimum relative entropy?**

$$g_P^\star \stackrel{?}{=} \mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m}).$$

# Kelly growth-rate is dual $\mathsf{D}_{\inf}$

For $P$ on $[0,1]$ with $\mathbb{E}_P[X] > m$, recall

$$\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m}) = \inf_{\substack{Q \in \mathcal{P}([0,1]) \\ \mathbb{E}_Q[X] \le m}} D(P \parallel Q).$$

This is an **infinite-dimensional convex program**, but with one inequality constraint

It turns out that its dual is precisely the **Kelly growth-rate maximization**:

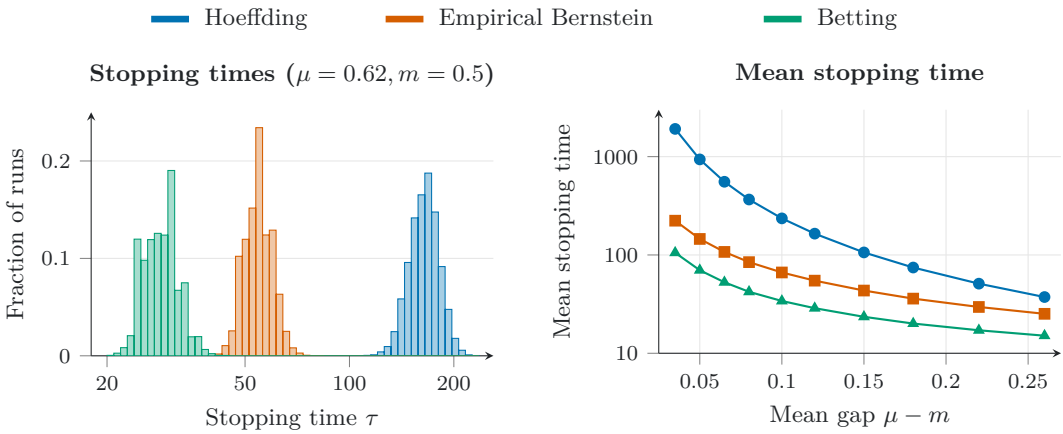
$$\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m}) = \sup_{u \in [0,1]} \mathbb{E}_P \left[ \log \left( 1 - u + u \frac{X}{m} \right) \right] := g_P^\star.$$

Combining this duality with our stopping-time bounds,

$$\lim_{\alpha \downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m})}.$$

# Empirical Comparison

**Baselines:** Tests based on time-uniform concentration inequalities

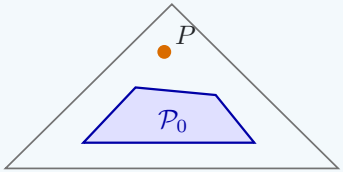


# Summary of the two approaches

**Finite alphabets**

Likelihood-based universal inference

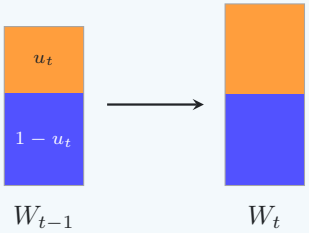
$$E_t = \frac{\text{KT predictor}}{\text{maximum null likelihood}}$$




**Bounded-mean testing**

Testing by betting

$$W_t = \prod_{i=1}^{t} \left(1 - u_i + u_i \frac{X_i}{m}\right)$$




**Is there a common design principle for a general testing problem?**

# A Unifying Design Principle

Likelihood-free Universal Inference

# Testing over general observation spaces

For some Polish space $\mathcal{X}$, let

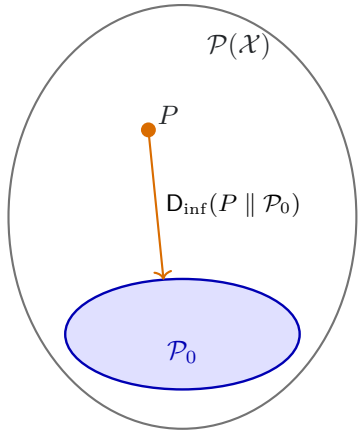
$$X_1, X_2, \ldots \overset{\text{iid}}{\sim} P, \qquad P \in \mathcal{P}(\mathcal{X}) \text{ unknown.}$$

Given $\mathcal{P}_0 \subseteq \mathcal{P}(\mathcal{X})$, we wish to test

$$H_0 : P \in \mathcal{P}_0 \qquad \text{vs.} \qquad H_1 : P \notin \mathcal{P}_0.$$

**Lower Bound:** For any level-$\alpha$, power-one test,

$$\mathbb{E}_P[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}, \quad \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) \coloneqq \inf_{Q \in \mathcal{P}_0} D(P\|Q).$$




**Can we construct tests attaining this limit without relying on likelihood functions?**

# Donsker–Varadhan (DV) Representation

For any $P, Q \in \mathcal{P}(\mathcal{X})$, recall the **Donsker–Varadhan variational representation**:

$$\mathsf{D}(P \parallel Q) = \sup_{f \in C_b(\mathcal{X})} \left\{ \mathbb{E}_P[f(X)] - \log \mathbb{E}_Q[e^{f(X)}] \right\},$$

where $C_b(\mathcal{X})$ denotes the space of bounded continuous functions on $\mathcal{X}$.

Interestingly, when $P \ll Q$ and $\log(dP/dQ) \in C_b(\mathcal{X})$, an optimizer is

$$f_Q^\star(x) = \log \frac{dP}{dQ}(x).$$

**Key observation:** The DV representation allows us to write the log-likelihood ratio as the solution of an optimization problem.

# Likelihood-Free UI for $\mathcal{P}_0 = \{Q\}$

For any predictable sequence $f_t \in C_b(\mathcal{X})$, define

$$M_0^Q = 1, \qquad M_t^Q = \prod_{i=1}^{t} \frac{e^{f_i(X_i)}}{\mathbb{E}_Q[e^{f_i(X)}]}.$$

$$X^{t-1} \xrightarrow{\text{Choose}} f_t \xrightarrow{\text{Observe}} X_t \xrightarrow{\text{Update}} M_t^Q$$

**Validity.** $\{M_t^Q\}$ is a nonnegative martingale under $Q$:

$$\mathbb{E}_Q[M_t^Q \mid \mathcal{F}_{t-1}] = M_{t-1}^Q \frac{\mathbb{E}_Q[e^{f_t(X)}]}{\mathbb{E}_Q[e^{f_t(X)}]} = M_{t-1}^Q.$$

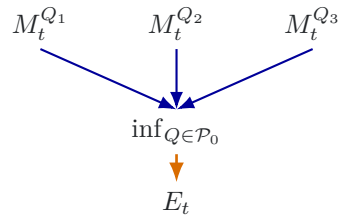
**Log-growth.** For a fixed function $f_t \equiv f$, under an i.i.d. alternative $P$,

$$\frac{1}{t} \log M_t^Q \quad = \quad \frac{1}{t} \sum_{i=1}^{t} f_t - \log \mathbb{E}_Q[e^{f_t}] \quad \xrightarrow{\text{a.s.}} \quad \mathbb{E}_P f - \log \mathbb{E}_Q e^f.$$

# Likelihood-Free UI for Composite $\mathcal{P}_0$

For each $Q \in \mathcal{P}_0$, construct $\{M_t^Q\}$ using predictable $\{f_{t,Q}\}$

$$M_t^Q = \prod_{i=1}^{t} \frac{e^{f_{i,Q}(X_i)}}{\mathbb{E}_Q[e^{f_{i,Q}(X)}]}.$$




As in Part 1, take the pointwise infimum:

$$E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q.$$

**Validity.** $\tau_\alpha = \inf\{t \geq 1 : E_t \geq 1/\alpha\}$ satisfies uniform level-$\alpha$ property under $\mathcal{P}_0$.

$E_t \leq M_t^{Q_0}$ Ville's ineq.

$$\sup_{Q_0 \in \mathcal{P}_0} \mathbb{P}_{Q_0}\left(\sup_{t \geq 0} E_t \geq \frac{1}{\alpha}\right) \leq \sup_{Q_0 \in \mathcal{P}_0} \mathbb{P}_{Q_0}\left(\sup_{t \geq 0} M_t^{Q_0} \geq \frac{1}{\alpha}\right) \leq \alpha.$$

# Summary of the Likelihood-Free UI Algorithm

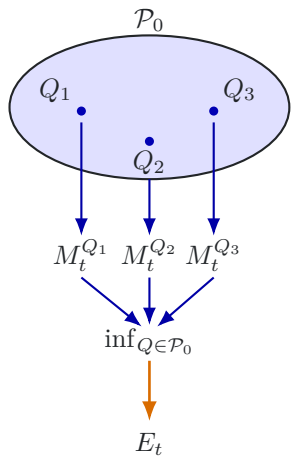
**Initialize:** $M_0^Q = 1$ for every $Q \in \mathcal{P}_0$.

**For** $t = 1, 2, \ldots$:

1. **Choose** predictable $f_{t,Q}$ for each $Q \in \mathcal{P}_0$.
2. **Observe** the next $X_t$.
3. **Update** each null-specific martingale

$$M_t^Q = M_{t-1}^Q \frac{e^{f_{t,Q}(X_t)}}{\mathbb{E}_Q[e^{f_{t,Q}(X)}]}.$$

4. **Aggregate.** $E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q$.
5. **Reject** $H_0$ if $E_t \geq 1/\alpha$.

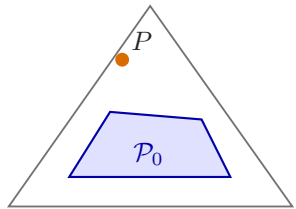


# Recovering finite-alphabet universal inference

Suppose $\mathcal{X} = \{1, \ldots, k\}$, and assume each null pmf $q$ is positive.

**Predictable function:** For each $t \geq 1$, choose

$$f_{t,Q}(x) = \log \frac{\widehat{p}_t(x)}{q(x)}, \quad \text{where} \quad \widehat{p}_t = \text{KT-predictor}$$




**Null normalization.** For every $Q$ with pmf $q$, we have

$$\mathbb{E}_Q[e^{f_{t,Q}(X)}] = \sum_{x \in \mathcal{X}} q(x) \frac{\widehat{p}_t(x)}{q(x)} = 1.$$

Substituting into likelihood-free UI gives

$$E_t = \inf_{Q \in \mathcal{P}_0} \prod_{i=1}^{t} \frac{\widehat{p}_i(X_i)}{q(X_i)} = \frac{\widetilde{P}(X^t)}{\sup_{Q \in \mathcal{P}_0} Q^t(X^t)}.$$

# Recovering bounded-mean betting

Let $X_t \in [0,1]$ and $\mathcal{P}_0 = \{Q : \mathbb{E}_Q X \le m\}$.

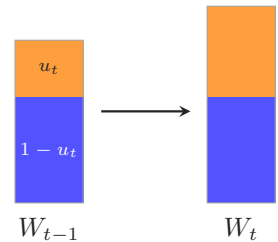
**Predictable function:** $f_{t,Q} = f_{u_t}$ for predictable $u_t$, where

$$f_u(x) = \log\left(1 - u + u\frac{x}{m}\right), 0 \le u < 1.$$

**Null normalization.** For every $Q$ with $\mathbb{E}_Q[X] \le m$,

$$\mathbb{E}_Q[e^{f_{t,Q}(X)}] = 1 - u_t + u_t\frac{\mathbb{E}_Q X}{m} \le 1,$$

with equality if $\mathbb{E}_Q X = m$ (e.g., $Q = \text{Bernoulli}(m)$).




On taking an infimum over $Q$, we have

$$E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q = \prod_{i=1}^{t}\left(1 - u_i + u_i\frac{X_i}{m}\right) = W_t.$$

# Beyond the two examples

**Two-sample testing**

Observe independent $X_t \sim P_X$ and $Y_t \sim P_Y$.

$$H_0 : \quad P_X = P_Y.$$

**Independence testing**

Observe i.i.d. pairs $(X_t, Y_t) \sim P_{XY}$.

$$H_0 : \quad P_{XY} = P_X \otimes P_Y.$$

**Invariance testing**

Observe $X_t \sim P_X$; let $G$ act on $\mathcal{X}$.

$$H_0 : \quad X \overset{d}{=} g(X), \quad \forall g \in G.$$

**Log-concavity testing**

Observe i.i.d. $X_t \sim P$ on $\mathbb{R}^d$, with density $p$.

$$H_0 : \quad p \text{ is log-concave.}$$

**The same likelihood-free UI idea applies to all these problems.**

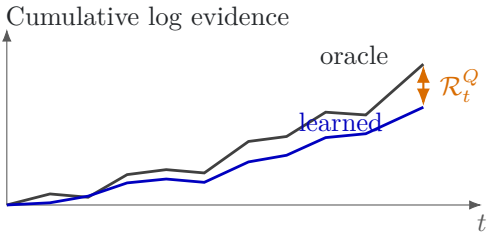
# The price of learning: regret

Fix an alternative $P \notin \mathcal{P}_0$. For each $Q \in \mathcal{P}_0$, compare:

- the **oracle** log likelihood process: $\Lambda_t^Q = \sum_{i=1}^t \log \frac{dP}{dQ}(X_i)$,
- the **learned** log evidence $\log M_t^Q$.

**Regret:** The gap between oracle and learned log evidence,

$$\mathcal{R}_t^Q := \Lambda_t^Q - \log M_t^Q.$$




**KT prediction:** regret w.r.t. to a fixed distribution.

**Universal portfolios:** regret w.r.t. to a fixed bet.

# Stronger regret control $\Longrightarrow$ stronger guarantees

Fix an alternative $P$ with $\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0) = \inf_{Q \in \mathcal{P}_0} \mathsf{D}(P \,\|\, Q) > 0$.

Let $\mathcal{R}_t = \sup_{Q \in \mathcal{P}_0} \mathcal{R}_t^Q$ be the *uniform regret* over the null distribution class.

| **Control on $\mathcal{R}_t$** | **Statistical Guarantee** | |
|---|---|---|
| $\limsup_{t \to \infty} \dfrac{\mathcal{R}_t}{t} < \mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)$ *a.s.* | $\mathbb{P}_P(\tau_\alpha < \infty) = 1$ | (Power one) |
| $\mathcal{R}_t \le \rho t$, $\rho < \mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)$, *w.h.p.* | $\mathbb{E}_P[\tau_\alpha] < \infty$ | (Finite expectation) |
| $\mathcal{R}_t \le r_t$ with $r_t = o(t)$, *w.h.p.* | $\mathbb{E}_P[\tau_\alpha] \sim \dfrac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)}$ | (First-order opt.) |

# Sufficient conditions for power one property

Under certain conditions, we can show that

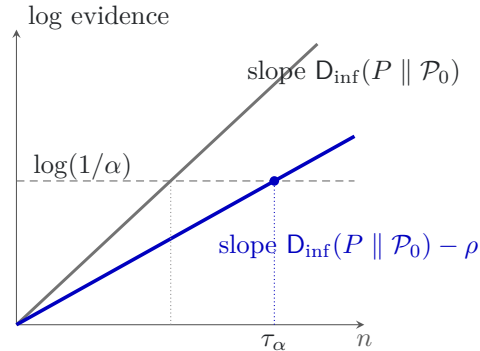
$$\frac{\log E_t}{t} \gtrsim \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) - \frac{\mathcal{R}_t}{t}.$$

Suppose $\limsup_{t\to\infty} \frac{\mathcal{R}_t}{t} \leq \rho < \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)$ a.s.

Then, the asymptotic growth rate is positive:

$$\liminf_{t\to\infty} \frac{\log E_t}{t} \geq \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) - \rho > 0.$$

**Hence,** $\{E_t : t \geq 1\}$ crosses $\log(1/\alpha)$ w.p. 1.

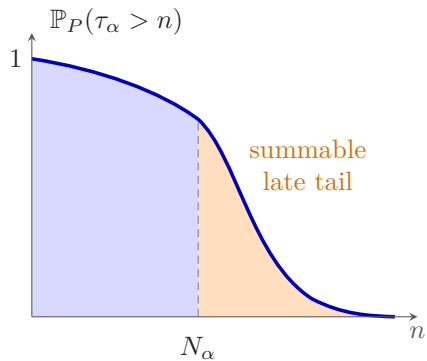


# From regret to expected stopping time

$$\mathbb{E}_P[\tau_\alpha] = \sum_{n\geq 0} \mathbb{P}_P(\tau_\alpha > n) \leq N_\alpha + \underbrace{\sum_{n\geq N_\alpha} \mathbb{P}_P(\tau_\alpha > n)}_{\text{Late-stopping tail}}.$$

**Finite mean:** A summable late-stopping tail for some $N_\alpha$ is sufficient.

**Optimality:** Choose $N_\alpha = \left\lceil \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P\|\mathcal{P}_0)-\varepsilon} \right\rceil$.




Under stronger regret control, can show

$$\mathbb{E}_P[\tau_\alpha] \leq N_\alpha + O_{P,\varepsilon}(1).$$

Letting $\alpha \downarrow 0$ and then $\varepsilon \downarrow 0$:

$$\lim_{\alpha\downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}.$$

# Summary of the general design principle

1. **Construct null-specific martingales.**
   For every $Q \in \mathcal{P}_0$, construct a nonnegative martingale $\{M_t^Q\}_{t\geq 0}$ under $Q$, using predictable witness functions $\{f_{t,Q} : t \geq 1\}$.

2. **Aggregate by pointwise minimization.**

$$E_t = \inf_{Q\in\mathcal{P}_0} M_t^Q.$$

   The resulting e-process is anytime-valid under every $Q \in \mathcal{P}_0$.

3. **Control regret to guarantee efficiency.**
   Stronger regret and tail guarantees yield stronger statistical performance:

$$\text{Power one} \quad \longrightarrow \quad \text{Finite } \mathbb{E}_P[\tau_\alpha] \quad \longrightarrow \quad \text{First-order optimality.}$$

# Some Extensions Beyond I.I.D. Testing

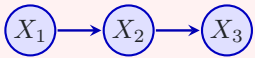
# Beyond the basic i.i.d. testing problem

Our basic i.i.d. testing framework has three ingredients:

| **Observation model** | **Inferential task** | **Performance criterion** |
|---|---|---|
| $X_t \overset{\text{i.i.d.}}{\sim} P$ | $H_0 : \ P \in \mathcal{P}_0$ | $\mathbb{E}_P[\tau_\alpha]$ |
| Fully observed, fixed $P$ | Reject or continue | Stop quickly on average |

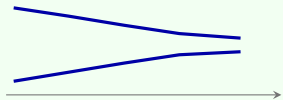
**Change the observations**

Markov chains; censored data



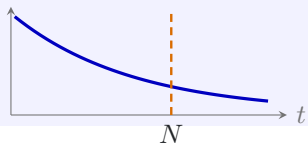

**Change the task**

Confidence sequences; change detection



**Change the objective**

Deadlines; time-sensitive rewards



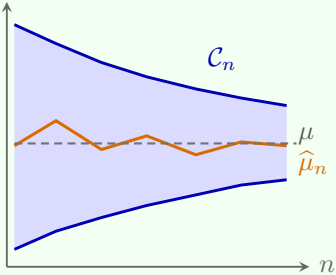
# Modified Task 1: Estimation

Suppose $X_1, X_2, \ldots \overset{\text{i.i.d.}}{\sim} P$, and we want to estimate the mean $\mu = \mathbb{E}_P[X]$.

**Formally,** this involves constructing intervals $\{\mathcal{C}_n\}_{n \geq 1}$, adapted to the data, such that

$$\mathbb{P}_P(\forall n \geq 1 : \ \mu \in \mathcal{C}_n) \geq 1 - \alpha.$$

This is a level-$(1-\alpha)$ confidence sequence (CS) for $\mu$




Contrast with fixed-time confidence intervals:

$$\underbrace{\mathbb{P}_P(\mu \in \mathcal{C}_n) \geq 1 - \alpha}_{\text{fixed } n} \qquad \text{vs.} \qquad \underbrace{\mathbb{P}_P(\forall n \geq 1 : \ \mu \in \mathcal{C}_n) \geq 1 - \alpha}_{\text{all times simultaneously}}.$$

The coverage guarantee holds for all times simultaneously; hence for all random stopping times as well.

# Constructing CSs by inverting anytime-valid tests

For each candidate mean $m \in [0,1]$, test the point null $H_{0,m}: \ \mu = m$ using an evidence process $\{M_n(m)\}_{n\geq 1}$.

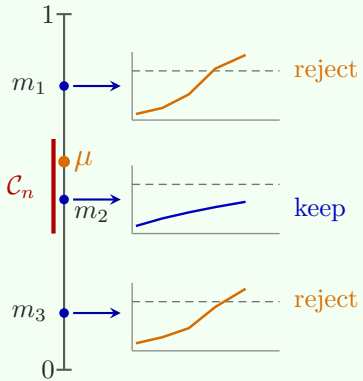
**Invert the tests.** Keep exactly those values that have not yet been rejected:

$$\mathcal{C}_n = \left\{ m : \ \max_{1\leq s\leq n} M_s(m) < \frac{1}{\alpha} \right\}.$$

At $m^\star = \mu$, we have by Ville's inequality,

$$\mathbb{P}_P(\exists n \geq 1 : \ m^\star \notin \mathcal{C}_n) = \mathbb{P}_P\left( \sup_{n\geq 1} M_n(m^\star) \geq \frac{1}{\alpha} \right) \leq \alpha.$$

We can also characterize the optimal size of $|\mathcal{C}_n|$ in terms of inverse $\mathsf{D}_{\mathsf{inf}}$ terms.



# Modified Observation Model 1: Markov chains

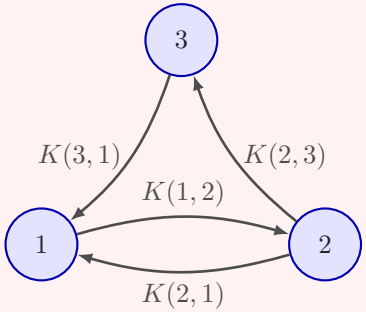
Consider a Markov chain on a finite state space $\mathcal{S} = \{1, \ldots, k\}$.

Let $K(\cdot, \cdot)$ denote the **unknown** transition matrix:

$$\mathbb{P}_K(X_t = y \mid \mathcal{F}_{t-1}) = K(X_{t-1}, y).$$

Testing problem

$$H_0 : \ K \in \mathcal{K}_0 \qquad \text{vs.} \qquad H_1 : \ K \notin \mathcal{K}_0.$$




**Goal:** Design a level-$\alpha$ power-one test $\tau$ for the above problem with small $\mathbb{E}_K[\tau]$.

# The lower bound for Markov chains

Consider an ergodic alternative $K$ with stationary distribution $\pi_K$

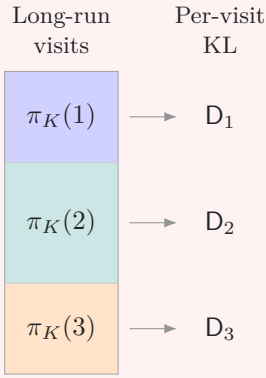
The relevant information quantity is

$$\mathsf{D}_{\inf}^{\mathrm{MC}}(K \parallel \mathcal{K}_0) := \inf_{L \in \mathcal{K}_0} \sum_{x \in \mathcal{S}} \pi_K(x)\, \mathsf{D}\big(K(x,\cdot) \parallel L(x,\cdot)\big).$$

**Lower bound.** Every level-$\alpha$, power-one test satisfies

$$\mathbb{E}_K[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}^{\mathrm{MC}}(K \parallel \mathcal{K}_0)} - O_K(1).$$

**Proof:** Combination of DPI and a modified Wald's identity.



# Universal inference for Markov Chains

**Finite-state Markov UI:** Replace symbol pmfs in the i.i.d. case by conditional transition probabilities

$$M_t^{\mathrm{MC}} = \frac{\prod_{i=2}^{t} \widehat{K}_i(X_i \mid X_{i-1})}{\sup_{L \in \mathcal{K}_0} \prod_{i=2}^{t} L(X_{i-1}, X_i)} = \frac{\text{Mixture of Conditional Likelihoods}}{\text{Maximum Null Conditional Likelihood}}.$$

**Stopping time:** $\tau_\alpha = \inf\{t \geq 2 : M_t^{\mathrm{MC}} \geq 1/\alpha\}$

**Unchanged:** Nullwise domination + Ville still guarantee anytime validity.

**New tools:** Universal predictors for Markov sources; Markov concentration; modified Wald's identity.

Under suitable conditions, $\mathbb{E}_K[\tau_\alpha] \sim \log(1/\alpha)/\mathsf{D}_{\inf}^{\mathrm{MC}}(K \parallel \mathcal{K}_0)$.

# Modified objective 1: Horizon-aware testing

So far, we have focused on minimizing $\mathbb{E}_P[\tau_\alpha]$.

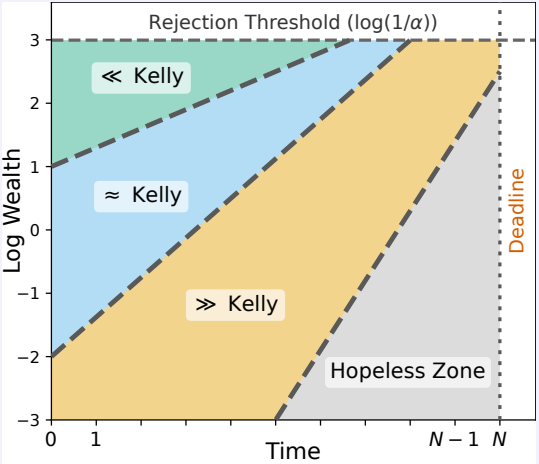
What if we have a **strict deadline** $N$?

**Horizon-aware objective.**

$$\begin{aligned} &\max \mathbb{P}_P(\tau_\alpha \le N), \\ \text{s.t.} \quad &\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha \le N) \le \alpha. \end{aligned}$$

**For the betting test**, it reduces to

$$\max_{\{u_t\}_{t=1}^N} \mathbb{P}_P\left( \max_{1 \le t \le N} \prod_{i=1}^{t} (1 - u_i + u_i \frac{X_i}{m}) \ge \frac{1}{\alpha} \right).$$



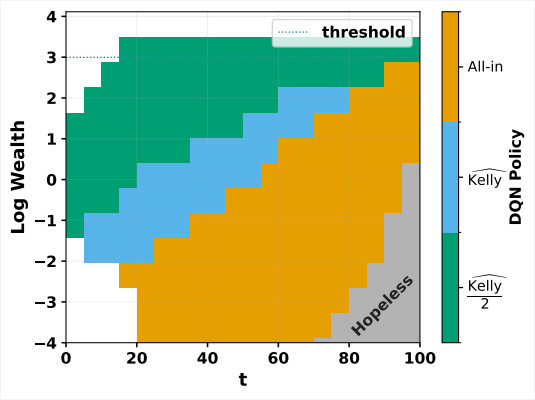
# Horizon-aware betting

**Finite-horizon control.** For a fixed alternative $P$, the state is $(t, \log W_t)$; the action is the next feasible bet.

**Theory.** Under suitable conditions:

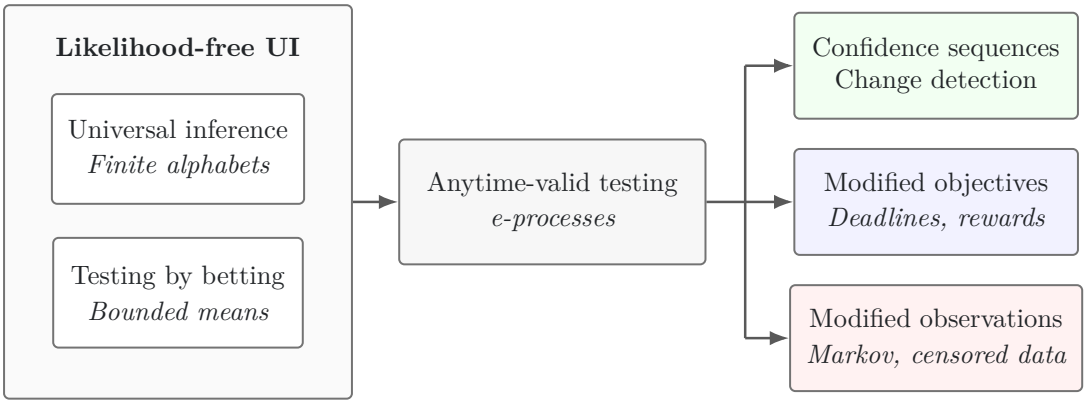
- **Ahead:** less aggressive than Kelly.
- **On schedule:** approximately Kelly.
- **Behind:** more aggressive than Kelly.

**Unknown $P$.** A Deep Q-Network (DQN) can learn to select bets using statistics of past observations.




Learned DQN actions.

# Summary so far



# Some Future Directions

# Beyond first-order optimality

**Our benchmark:** for every fixed alternative $P$,

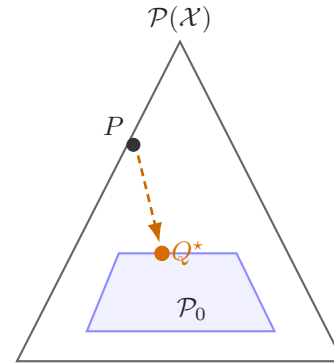
$$\mathbb{E}_P[\tau_\alpha] \sim \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)} \qquad (\alpha \downarrow 0).$$

**Second-order behavior.**

How does the stopping time fluctuate around its leading term?

**Time-uniform local detection boundary.**

What are the closest alternatives that can be reliably detected by a single anytime-valid test across all $n \in \mathbb{N}$?



# Active anytime-valid inference

**Multiple data sources.** Suppose $K$ sources are governed by a common hypothesis.

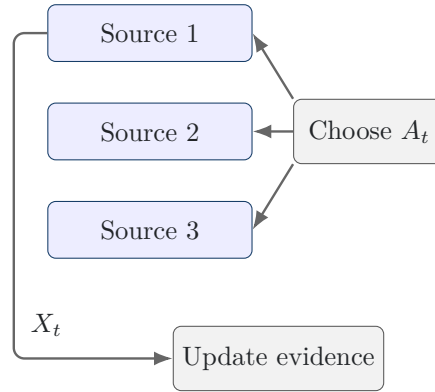
At each time $t$, the analyst chooses

$$A_t \in \{1, \ldots, K\}, \qquad X_t \sim P_{A_t},$$

using the observations collected so far.

Different sources may provide different amounts of information.

**Question.** What is the optimal joint policy for *sensing and stopping*?

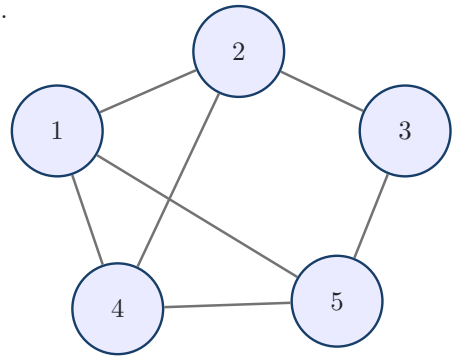


# Distributed anytime-valid inference

**Distributed observations:** multiple agents collect local data, satisfying a common underlying hypothesis.

Each agent maintains local evidence $E_t^{(i)}$ and communicates with its neighbors (or a central server).

**Question:** How can a network aggregate evidence with both validity and optimal information growth?

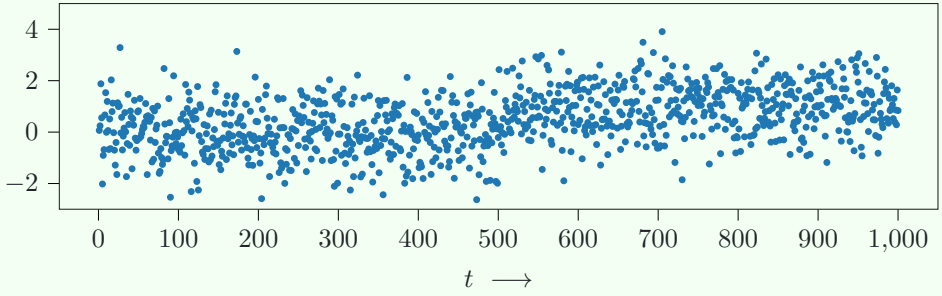


Thank you.

# Modified Task 2: Sequential change detection

Suppose the distribution of the observations changes at an unknown time $\nu$:

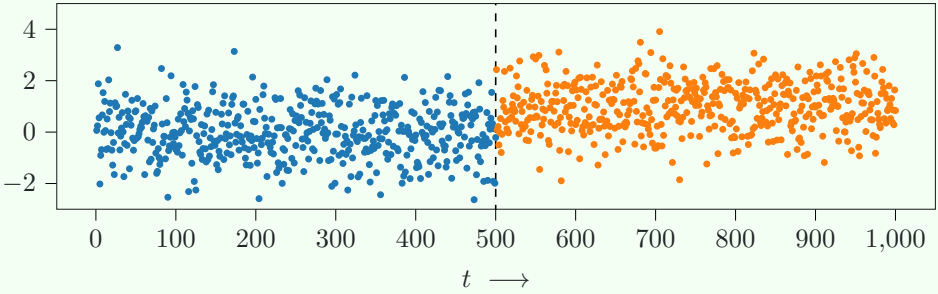
$$X_t \sim \begin{cases} P_{\text{pre}}, & t \leq \nu, \\ P_{\text{post}} \neq P_{\text{pre}}, & t > \nu. \end{cases}$$



# Modified Task 2: Sequential change detection

Suppose the distribution of the observations changes at an unknown time $\nu$:

$$X_t \sim \begin{cases} P_{\text{pre}}, & t \le \nu, \\ P_{\text{post}} \neq P_{\text{pre}}, & t > \nu. \end{cases}$$

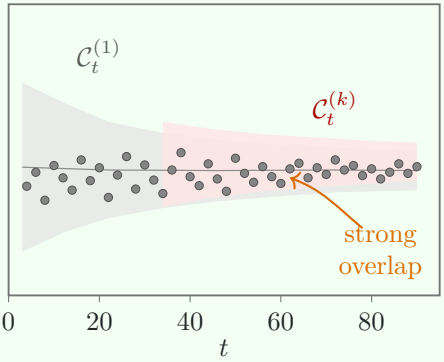



**Goal:** Detect the change quickly while keeping false alarms rare.

# Change detection using confidence sequences

1. At each round $k$, start a new CS $\{\mathcal{C}_t^{(k)} : t \geq k\}$.
2. At time $t$, compare all active confidence sequences $\mathcal{C}_t^{(1)}, \mathcal{C}_t^{(2)}, \ldots, \mathcal{C}_t^{(t)}$.
3. Stop as soon as they *disagree*, i.e.

$$\tau = \inf\Big\{t \geq 1 : \bigcap_{k=1}^{t} \mathcal{C}_t^{(k)} = \varnothing\Big\}.$$

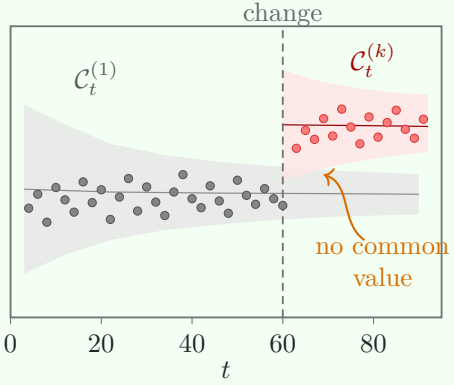



- When $\nu = \infty$, we have $\mathbb{E}_\infty[\tau] \geq \frac{1}{\alpha}$.
- When $\nu < \infty$, detection delay depends on quality of the CSs used.

# Change detection using confidence sequences

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3. Stop as soon as they *disagree*, i.e.

$$\tau = \inf\Big\{t \geq 1 : \bigcap_{k=1}^{t} \mathcal{C}_t^{(k)} = \varnothing\Big\}.$$



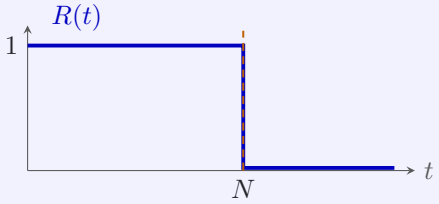

- When $\nu = \infty$, we have $\mathbb{E}_\infty[\tau] \geq \frac{1}{\alpha}$.
- When $\nu < \infty$, detection delay depends on quality of the CSs used.

# Modified objective 2: Time-sensitive testing

A deadline is one way to reward early detection. More generally, assign a nonnegative, nonincreasing reward $R(t)$ to rejection at time $t$.

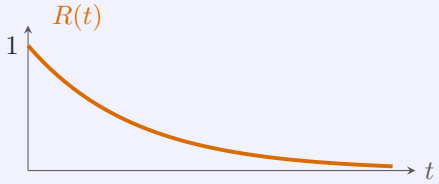
$$\sup_{\tau_\alpha \in \mathcal{T}_\alpha} \mathbb{E}_P[R(\tau_\alpha)], \qquad R(\infty) = 0,$$

**Hard deadline**




$R(t) = \mathbf{1}\{t \le N\}$ gives $\mathbb{P}_P(\tau_\alpha \le N)$.

**Discounted discovery**




$R(t) = e^{-ct}$ gives $\mathbb{E}_P[e^{-c\tau_\alpha}]$.

# Rényi divergences and reward decay

**Converse (ongoing work).** For a finite-alphabet alternative $P \notin \mathcal{P}_0$, let $\mathcal{I}_r(P) := \inf_{Q \in \mathcal{P}_0} D_r(P \,\|\, Q)$.
Write $\Phi_P(\eta) := \eta \mathcal{I}_{1/(1-\eta)}(P)$, where $0 < \eta < 1$. Every level-$\alpha$ test satisfies

$$\mathbb{E}_P[R(\tau_\alpha)] \;\le\; \inf_{0<\eta<1} \left\{ \alpha^\eta \sup_{n \ge 1} R(n) e^{n\Phi_P(\eta)} \right\}.$$

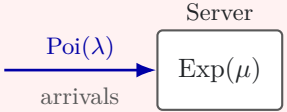
**Three regimes, depending on how rapidly the reward decays:**

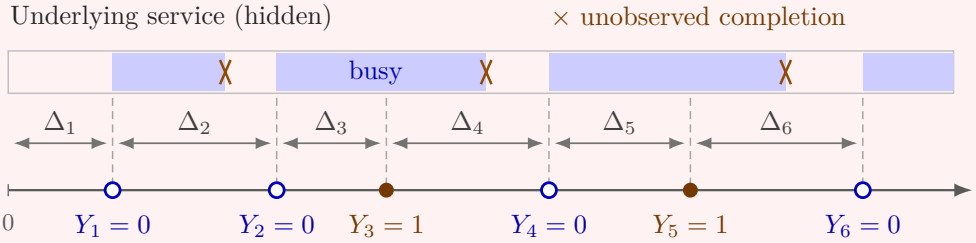
| | |
|---|---|
| **Subexponential**<br>$R(n) = e^{-o(n)}$ | KL / typical stopping time $\log(1/\alpha)/\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)$; a separate argument is needed. |
| **Exponential**<br>$R(n) = e^{-cn}$ | For $0 < c < \mathcal{I}_\infty(P)$, the Rényi exponent $\eta_c$ solves $\Phi_P(\eta_c) = c$. |
| **Superexponential**<br>$R(n) \lesssim e^{-cn}$ | If $c \ge \mathcal{I}_\infty(P)$, the reward is $O(\alpha)$; randomized early rejection attains this order. |

Here $\mathcal{I}_\infty(P) := \inf_{Q \in \mathcal{P}_0} D_\infty(P \,\|\, Q)$.

# Modified Observations 2: Censoring

- A single server with service times $S \sim \text{Exp}(\mu)$.
- Jobs arrive as a Poisson process (rate $\lambda$).
- An arriving job is admitted whenever the server is idle.




**Observations:** inter-arrival times $\{\Delta_t : t \geq 1\}$ and busy/idle indicators $\{Y_t : t \geq 1\}$.

**Goal:** For some known $\mu^\star > 0$, decide between $H_0 : \ \mu \geq \mu^\star$ vs. $H_1 : \ \mu < \mu^\star$.

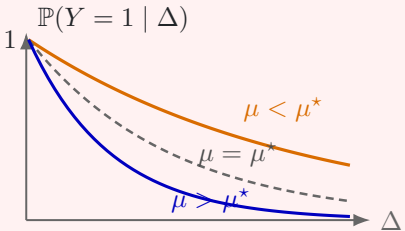
# Anytime-Valid Test From Occupancy Indicators

**Challenge:** We want to test a property of service times $\{S_t : t \geq 1\}$, but we don't directly observe them!

Under the Poisson-arrival assumption: $\Delta_t \sim \mathrm{Exp}(\lambda)$.

Under the additional Exponential service-time assumption,

$$Y_t \mid (\Delta_t, \mathcal{F}_{t-1}) \sim \mathrm{Ber}(e^{-\mu \Delta_t}).$$




An instantiation of the likelihood-based universal inference procedure gives, for fixed $\mu < \mu^\star$,

$$\mathbb{E}_{\mu,\lambda}[\tau_\alpha] \sim \frac{\log(1/\alpha)}{I_{\mu,\lambda}}, \quad \text{where} \quad I_{\mu,\lambda} = \mathbb{E}_\Delta\left[\mathsf{d}\big(e^{-\mu\Delta}, e^{-\mu^\star \Delta}\big)\right],$$

Similar ideas extend to more general service time distributions.